

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

B.Tech Degree S5 (R, S) / S5 (WP) (R) / S3 (PT) (S,FE) Examination November 2024 (2019 Scheme)

**Course Code: EET 307****Course Name: SYNCHRONOUS AND INDUCTION MACHINES**

Max. Marks: 100

Duration: 3 Hours

**(GRAPH SHEET NEEDED)****PART A***(Answer all questions; each question carries 3 marks)*

Marks

- |    |                                                                                                                                               |   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1  | Explain the effect of harmonics in pitch factor and distribution factor.                                                                      | 3 |
| 2  | Sketch the phasor diagram of a cylindrical pole type alternator with lagging and leading power factor load.                                   | 3 |
| 3  | $X_d$ and $X_q$ are different for salient pole alternators where as they are same for cylindrical rotor alternators. Comment on the statement | 3 |
| 4  | Identify the voltage regulation method which is known as the optimistic method and explain why it is known so.                                | 3 |
| 5  | Enlist the starting methods of synchronous motor and describe any one method.                                                                 | 3 |
| 6  | Sketch the torque slip characteristics of three phase induction motor and show the variation of rotor resistance in the characteristics.      | 3 |
| 7  | Explain the phenomenon of crawling in three phase induction motor                                                                             | 3 |
| 8  | Describe the pole-changing method for speed control of the induction motor.                                                                   | 3 |
| 9  | Compare synchronous generator and induction generator.                                                                                        | 3 |
| 10 | Illustrate how starting torque is developed in split phase induction motor.                                                                   | 3 |

**PART B***(Answer one full question from each module, each question carries 14 marks)***Module -1**

- |    |                                                                                                                                                                                                                                                            |   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 11 | a) Explain the advantages of stationary armature over rotating armature types                                                                                                                                                                              | 6 |
|    | b) A 4-pole, 50 Hz star-connected alternator has a flux per pole of 0.12 Wb. It has 4 slots per pole per phase, conductors per slot being 4. If the winding coil span is $150^\circ$ , calculate open circuit line voltage and open circuit phase voltage. | 8 |
| 12 | a) Derive the emf equation of alternator.                                                                                                                                                                                                                  | 5 |

### 1100EET307112401

- b) An 8-pole, 3-phase,  $60^\circ$  spread, double layer winding has 72 coils in 72 slots. 9  
The coils are short pitched by two slots. Calculate the winding factor for the fundamental and third harmonic.

#### Module -2

- 13 a) A 3-phase, star-connected, 1000 kVA, 11000 V alternator has rated current of 8  
52.5 A. The ac resistance of the winding per phase is  $0.45 \Omega$ .  
The test results are given below:  
OC test: Field current = 12.5A, Voltage between lines = 411V  
SC test: Field current = 12.5A, Line current = 52.5A  
Determine the full-load voltage regulation of the alternator at (a) 0.8 pf lagging  
and (b) 0.8 pf leading.
- b) Explain the slip test of alternator. 6
- 14 A 3.5 MVA, Y connected alternator rated at 4160 V at 50 Hz has open circuit 14  
characteristics given by the following data:

|                    |      |      |      |      |      |      |      |      |
|--------------------|------|------|------|------|------|------|------|------|
| Field current (A): | 50   | 100  | 150  | 200  | 250  | 300  | 350  | 400  |
| Line EMF (V)       | 1620 | 3150 | 4160 | 4750 | 5130 | 5370 | 5550 | 5650 |

A field current of 200 A is found necessary to circulate full load current on short circuit of the alternator. Calculate by (i) Synchronous Impedance method and (ii) Ampere-turn method, the full load voltage regulation at 0.8 pf lagging. Neglect armature resistance.

#### Module -3

- 15 a) Synchronous motor is not self starting. Justify the statement. 6  
b) A 440 V, 50 Hz, 6-pole, 3 phase induction motor draws an input power of 76 kW 8  
from the mains. The rotor emf makes 120 complete cycles per minute. Its stator losses are 1 kW and rotor current per phase is 62 A. Calculate (a) speed (b) rotor copper losses (c) rotor resistance per phase (d) torque developed.
- 16 a) Sketch the power angle characteristics of synchronous machine and explain. 6  
b) The power input to a 6-pole, 50 Hz, 3-phase induction motor is 700 W at no-load 8  
and 10 kW at full-load. The no-load copper losses may be assumed negligible

### 1100EET307112401

while the full-load stator and rotor copper losses are 295 W and 310 W respectively. Calculate the full-load speed, shaft torque and efficiency of the motor assuming rotational and core losses to be equal.

#### Module -4

- 17 a) A 30hp, 500 V, 50 Hz, 4 pole, 3 phase mesh connected induction motor has the following data: 14
- (i) No load test: 500 V, 8.3 A, 1500 W
  - (ii) SC test :100 V, 32 A, 1600 W
  - (iii) Rotor copper loss at standstill is equal to half of the total copper loss
- Sketch the circle diagram and estimate the full load current and power factor.
- 18 a) Elaborate the working of Double cage induction motor with relevant sketches. Also draw the torque slip characteristics of the Double cage induction motor 9
- b) Explain the dynamic braking of the Induction motor. 5

#### Module -5

- 19 a) Describe with neat figure, the operation of Induction Generator 8
- b) Why Single phase induction motors are not self-starting? 6
- 20 a) Explain with neat sketches about the different types of single-phase induction motors with their applications. 14

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